SACHIN RAJPUT

Software Development Engineer (SDE) Intern | Computer Science Student | Class of 2028

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UG 3rd Year, Computer Engineering | SGSITS, Indore | Enrollment No. 0801CS241119

SUMMARY

Computer Science undergraduate (graduating 2028) with strong fundamentals in object-oriented design, data structures, algorithm development, and complexity analysis, backed by 250+ problems solved on LeetCode. Proficient in Java, C++, and Python, with working knowledge of HTML, CSS, JavaScript, and React.js, plus coursework in SQL and relational database concepts. A fast learner who thrives in fast-paced, ambiguous problem spaces, self-starts on new technologies, and communicates technical challenges and solutions clearly. Seeking a Software Development Engineer (SDE) Internship to apply strong analytical and problem-solving skills to real-world engineering problems.

TECHNICAL SKILLS

- Programming Languages: Java, C++, Python, C
- Computer Science Fundamentals: Object-Oriented Design (OOD), Data Structures & Algorithms, Algorithm Development, Time & Space Complexity Analysis
- Data Structures & Algorithms: Arrays, Strings, Linked Lists, Stacks, Queues, Trees, Tries, Binary Search, Hashing
- Databases: SQL, Relational Database Concepts, MySQL (Joins, Subqueries, Aggregation, Window Functions)
- Web Development: HTML5, CSS3, JavaScript (ES6+), React.js, REST API Integration, Responsive Web Design
- Version Control & Tools: Git, GitHub, VS Code, Vite, npm, Vercel
- Ways of Working: Agile Methodology, Self-Directed Learning, Problem-Solving in Ambiguous/Undefined Scenarios

PROJECTS

Prefix Search Engine | Trie Data Structure Implementation

- Designed and engineered a high-performance, array-based Trie (Prefix Tree) to implement search-as-you-type and autocomplete functionality, applying strong data structures and algorithmic thinking.
- Achieved O(L) time complexity for search and insert operations through careful complexity analysis, optimizing performance over standard linear search methods.
- Implemented memory-efficient nodes to handle large datasets, independently navigating design trade-offs to scale the system with growing dictionaries.

React Chatbot | Interactive Chat Application

- Built a single-page application in React.js (Vite) using object-oriented, component-based architecture with separate input, message-list, and message components.
- Implemented controlled input handling, keyboard event support, and state management using React hooks (useState, useRef, useEffect).
- Independently learned and applied a new library (dayjs) to solve a formatting requirement not covered by the base project scope.

Currency Converter | Real-Time Data Application

- Developed a currency converter using HTML5, CSS3, and JavaScript, integrating a REST API via the Fetch API and async/await for real-time exchange rate data.
- Implemented dynamic input handling and validation to compute and render accurate results instantly.

GitHub: Currency-Converter-Project

JavaScript Mini Games | Tic Tac Toe & Stone Paper Scissors

- Built two browser-based games using HTML5, CSS3, and JavaScript, implementing game-state management, event handling, and win/draw detection logic.

GitHub: Tic-Tac-Toe-Project | Rock-Paper-Scissors-Game

EDUCATION

B.Tech, Computer Engineering — Shri G.S. Institute of Technology and Science (SGSITS), Indore (R.G.P.V.)

Expected Graduation: 2028 | CGPA: 8.20

12th (M.P.B.S.E) — Modal Foundation H.S. School, Gwalior — 2024 — 69.6%

10th (M.P.B.S.E) — Gouri Shankar H.S. School, Gwalior — 2022 — 89.3%

ACHIEVEMENTS

- Secured All India Rank 22777 in JEE-Main out of 1.4 million candidates (2024).
- Solved 250+ Data Structures & Algorithms problems on LeetCode, covering Arrays, Strings, Linked Lists, Stacks, Queues, Trees, Binary Search, and Hashing — demonstrating consistent, self-directed skill-building.
- Secured 2nd Rank at School Science Fair for designing and prototyping a low-cost mechanical prosthetic hand among 50+ participants (2020).

CERTIFICATIONS

- SQL (Basic & Intermediate) — HackerRank: joins, subqueries, aggregation, and window functions.
- Pandas — Kaggle: data structures, indexing, filtering, grouping, and merging.
- Python Programming Fundamentals — Kaggle: data types, control flow, functions, and Object-Oriented Programming.

RELEVANT COURSEWORK

Data Structures and Algorithms, Object-Oriented Programming, Design and Analysis of Algorithms, Database Management Systems (SQL & Relational Databases), Machine Learning, Data Science and Engineering, Internet of Things, Agile Methodology